

Problem Sheet 5: Solutions

Answer 1.

a) $X_1(j\omega) = 2\pi\delta(\omega) + \pi\delta(\omega - 4\pi) + \pi\delta(\omega + 4\pi)$

$$X_1(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X_1(j\omega) e^{j\omega t} d\omega$$

$$X_1(t) = 1 + \frac{1}{2} (e^{j4\pi t} + e^{-j4\pi t}) = 1 + \cos 4\pi t$$

b) $X_2(j\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X_2(j\omega) e^{j\omega t} d\omega$

$$X_2(j\omega) = \frac{1}{2\pi} \left[\int_{-2}^0 -2e^{j\omega t} d\omega + \int_0^2 2e^{j\omega t} d\omega \right]$$

$$X_2(j\omega) = \frac{2}{j\pi t} [\cos 2t - 1]$$

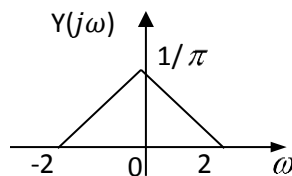
Answer 2.

a) Let $f(t) = \frac{\sin t}{\pi t}$

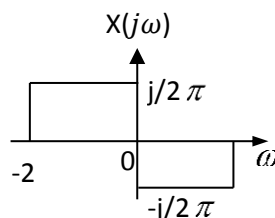
Fourier transform of $f(t)$ is $F(j\omega)$

Now multiply in time domain

$$y(t) = f(t) \times f(t) \leftrightarrow \frac{F(\omega) * F(\omega)}{2\pi}$$



Fourier transform of $t[f(t)]^2$ is $j \frac{d}{d\omega} Y(\omega)$



$$\text{b) } \int_{-\infty}^{\infty} t^2 \left(\sin \frac{t}{\pi t} \right)^4 = \frac{1}{2\pi} \int X(j\omega)^2 d\omega = \frac{1}{2\pi} \times \frac{1}{\pi^2} = \frac{1}{2\pi^3}$$

Answer 3.

$$\text{Given: } Y(\omega) = X(\omega)H(\omega)$$

$$\text{W.K.T } F\{x(at)\} = \frac{1}{a} X\left(\frac{\omega}{a}\right)$$

$$G(\omega) = \frac{1}{9} X\left(\frac{\omega}{3}\right) H\left(\frac{\omega}{3}\right)$$

$$G(\omega) = \frac{1}{9} Y\left(\frac{\omega}{3}\right)$$

$$G(\omega) = \frac{1}{3} \times \frac{1}{3} Y\left(\frac{\omega}{3}\right)$$

Take Fourier Inverse on both sides:

$$g(t) = \frac{1}{3} y(3t)$$

$$\text{Thus } A = \frac{1}{3} \text{ and } B = 3$$

Answer 4.

$$F^{-1}\{(1 + j\omega)X(j\omega)\} = Ae^{-2t}u(t)$$

By taking Fourier transform both sides

$$(1 + j\omega)X(j\omega) = \frac{A}{2 + j\omega}$$

$$X(j\omega) = A \left(\frac{1}{1 + j\omega} - \frac{1}{2 + j\omega} \right)$$

Taking inverse Fourier transform

$$x(t) = Ae^{-t}u(t) - Ae^{-2t}u(t)$$

Using fact c)

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = 1$$

$$\int_{-\infty}^{\infty} A^2(e^{-2t} + e^{-4t} - 2e^{-3t})u(t)dt = 1$$

By solving and using the fact that $x(t)$ is non-negative, $A = \sqrt{12}$

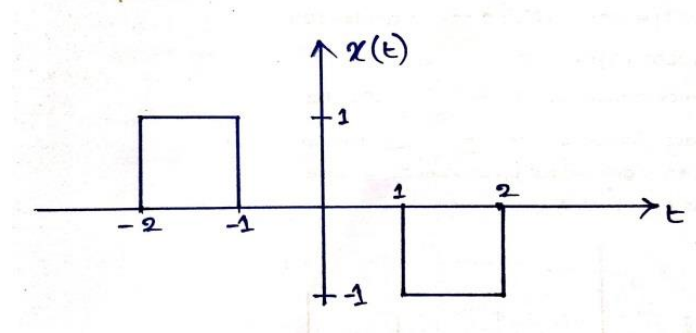
Answer 5.

a.)

1. Fourier transform is purely imaginary, so $x(t)$ is a real and odd. Therefore, signals in figures (a) and (d) have this property.
2. $x(t)$ is real and even. Therefore, signals in figures (e) and (f) have this property.
3. For there to exist a real ' α ', such that $e^{j\alpha\omega}X(j\omega)$ is real, we require that $x(t - \alpha)$ be a real or real and even signal. Therefore, signal in figures (a), (b), (e), and (f) have this property.
4. For this condition to be true, $x(t = 0) = 0$. Therefore, signal in figures (a), (b), (c), (d), and (f) have this property.
5. For this condition to be true, the derivative of $x(t)$ has to be zero at $t = 0$. Therefore, signals in figures (b), (c), (e), and (f) have this property.
6. The signals in figure (a) and (b) have this property. Figure (a) has this property because it is the integration of shifted square wave. The Fourier transform of the signal that shown in Figure (b) is $X(j\omega) = 2e^{-j\omega}$ which is periodic.

b.)

The signal with only properties 1, 4 and 5



Answer 6:

a.)

$$g(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} G(j\omega) e^{j\omega t} d\omega = \frac{\sin 2t}{\pi t} = x(t) \cos(t)$$

$$x(t) = \frac{2\sin(t)}{\pi t}$$

b.)

$$x_1(t) = \frac{g(t)}{\cos \frac{2}{3}t} = \frac{\sin 2t}{\pi t \cos \frac{2}{3}t}$$

Answer 7:

a.) Using the multiplication property of Fourier transform (i.e. multiplication in time domain implies convolution in Fourier domain, we get):

$$H(j\omega) = \frac{1}{2\pi} H(j\omega) * U(j\omega)$$

$$\Rightarrow H(j\omega) = \frac{1}{2\pi} H(j\omega) * \left(\frac{1}{j\omega} + \pi\delta(\omega) \right)$$

$$\Rightarrow H(j\omega) = \frac{1}{2} H(j\omega) + \frac{1}{2\pi j} (H(j\omega) * \frac{1}{\omega})$$

$$\Rightarrow H(j\omega) = \frac{1}{\pi j} \left(H(j\omega) * \frac{1}{\omega} \right)$$

$$\Rightarrow H(j\omega) = \frac{1}{\pi j} \int_{-\infty}^{\infty} \frac{H(j\tau)}{\omega - \tau} d\tau$$

$$\Rightarrow H_R(j\omega) + jH_I(j\omega) = \frac{1}{\pi j} \int_{-\infty}^{\infty} \frac{H_R(j\tau) + jH_I(j\tau)}{\omega - \tau} d\tau = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{H_I(j\tau) - jH_R(j\tau)}{\omega - \tau} d\tau$$

$$\Rightarrow H_R(j\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{H_I(j\tau)}{\omega - \tau} d\tau \text{ and } H_I(j\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{H_R(j\tau)}{\omega - \tau} d\tau$$

b.)

$$y(t) = x(t) * \frac{1}{\pi t}$$

Using linearity property of Fourier transform, we get:

$$u(t) \xleftrightarrow{FT} \frac{1}{j\omega} + \pi\delta(\omega)$$

$$2u(t) - 1 \xleftrightarrow{FT} \frac{2}{j\omega}$$

Using duality of Fourier transform, we get:

$$\frac{2}{jt} \stackrel{FT}{\leftrightarrow} 2\pi[2u(-w) - 1]$$

$$\frac{1}{\pi t} \stackrel{FT}{\leftrightarrow} j[2u(-w) - 1]$$

$$\Rightarrow FT\left(\frac{1}{\pi t}\right) = j[2u(-w) - 1]$$

$$Y(jw) = X(jw)H(jw)$$

$$H(jw) = j[2u(-w) - 1] = \begin{cases} -j, w > 0 \\ j, w < 0 \end{cases}$$

c.)

Let the Hilbert transform of $x(t)$ be $y(t)$

$$Y(jw) = X(jw)H(jw) = \pi[\delta(w - 3) + \delta(w + 3)]H(jw) = j\pi\delta(w + 3) - j\pi\delta(w - 3)$$

$$\Rightarrow y(t) = \sin(3t)$$

Answer 8.

a.) Since $H(j\omega)$ is real and even $h(t)$ is also real and even.

As we have $H(j\omega) = H(-j\omega) \Rightarrow h(t) = h(-t)$ and $H(j\omega) = H(j\omega)^* \Rightarrow h(t) = h(-t)^*$

$$|h(t)| = \left\| \frac{1}{2\pi} \int_{-\infty}^{\infty} H(jw)e^{i\omega t} dw \right\| \leq \frac{1}{2\pi} \int_{-\infty}^{\infty} |H(jw)e^{i\omega t}| dw \text{ and since } |e^{j\omega t}| < 1$$

$$\text{Hence } |h(t)| \leq \frac{1}{2\pi} \int_{-\infty}^{\infty} |H(w)| dw = h(0)$$

b.) B.W is 2ω

c.) $B_w H(j0) = \text{Area under } H(j\omega)$

$$B_w = \frac{1}{H(j0)} \int_{-\infty}^{\infty} H(jw) dw.$$

$$\text{d.) } t_r = \frac{s(\infty)}{h(0)} = \frac{\int_{-\infty}^{\infty} h(t) dt}{\frac{1}{2\pi} \int_{-\infty}^{\infty} H(jw) dw} = \frac{H(j0)}{\frac{1}{2\pi} \int_{-\infty}^{\infty} H(jw) dw} = \frac{2\pi}{\text{Area_under_}H(jw)} = \frac{2\pi}{B_w}$$

e.) Simply use the result obtained in part d)

Answer 9.

a.) A raised cosine pulse in time domain is given by:

$$h(t) = \text{sinc}\left(\frac{t}{T}\right) \times \frac{\cos\left(\frac{\alpha\pi t}{T}\right)}{1 - \frac{4\alpha^2\alpha t^2}{T^2}}$$

Now we can consider $h(t)$ as a product of $h_1(t)$ and $h_2(t)$

For $h_1(t)$ the Fourier transform is given by:

$$H_1(w) = \frac{1}{T} \text{rect}\left(\frac{w}{\pi T}\right)$$

Where $\text{rect}(t) = u(t+1) - u(t)$

Based on duality

And for $h_2(t)$ we can re-write it as

$$\begin{aligned} \left(\frac{T}{2\alpha}\right)^2 \times \frac{\cos\left(\frac{\pi\alpha t}{T}\right)}{\left(1 - \left(\frac{2\alpha t}{T}\right)^2\right)} \quad \text{Now put } \tau = \frac{\alpha\pi}{T} \\ = \frac{\pi}{4\tau} \times \frac{\frac{\pi}{\tau} \cos(t\tau)}{\left(\left(\frac{\pi}{2\tau}\right)^2 - t^2\right)} \end{aligned}$$

Now we have that $\cos\left(\frac{\pi t}{2\tau}\right) \text{rect}\left(\frac{t}{2\tau}\right) \rightarrow \frac{\frac{\pi}{\tau} \cos(w\tau)}{\left(\left(\frac{\pi}{2\tau}\right)^2 - w^2\right)}$

So, by duality since $X(t) \rightarrow 2\pi x(-w)$

Hence the Fourier transform of $h_2(t)$ is $\frac{\pi}{4\tau} \cos\left(\frac{\pi w}{2\tau}\right) \text{rect}\left(\frac{w}{2\tau}\right)$ since both cosine and rect are even functions.

Hence $H(w) = H_1(w) * H_2(w)$

$$\Rightarrow \frac{1}{T} \text{rect}\left(\frac{w}{\pi T}\right) * \frac{\pi}{4\tau} \cos\left(\frac{\pi w}{2\tau}\right) \text{rect}\left(\frac{w}{2\tau}\right)$$

Which is the Fourier transform.

b.) $X(\omega) = \int_{-\infty}^{\infty} e^{-at^2} e^{-j\omega t} dt$

Making perfect square:

$$X(w) = \int_{-\infty}^{\infty} (e^{-a(t+\frac{jw}{2a})^2} dt) e^{-\frac{w^2}{4a}}$$

Taking $t' = \sqrt{a}\left(t + \frac{jw}{2a}\right)$ and $\sqrt{a}dt = dt'$

$$\Rightarrow \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} e^{-(t')^2} dt' e^{-\frac{w^2}{4a}} = \sqrt{\frac{\pi}{a}} e^{-\frac{w^2}{4a}}$$